

Problem 1

To solve the **Water Jug Problem** using **state space search** in order to measure **exactly 4 liters of water** using two jugs of capacities **3 liters** and **5 liters**.

Sample Input

Enter capacity of Jug A: 3
Enter capacity of Jug B: 5
Enter target amount: 4

Sample Output

Initial State: (0, 0)

Step 1: Fill Jug B
State: (0, 5)

Step 2: Pour water from Jug B to Jug A
State: (3, 2)

Step 3: Empty Jug A
State: (0, 2)

Step 4: Pour water from Jug B to Jug A
State: (2, 0)

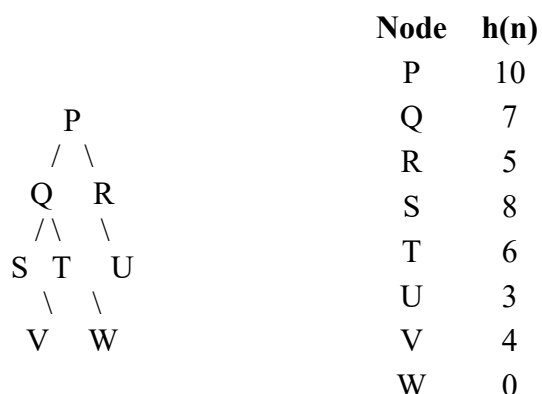
Step 5: Fill Jug B
State: (2, 5)

Step 6: Pour water from Jug B to Jug A
State: (3, 4)

Goal State Reached!
4 liters measured in Jug B.

Problem 2

To apply **Best First Search** to determine an optimal path between a start node and a goal node using **heuristic estimates**.



Goal Node **W**
Output- Start Node: P
Goal Node: W

Node Expanded: P
Node Expanded: R
Node Expanded: U
Node Expanded: W

Goal Node Reached Successfully!
Optimal Path Found:
P → R → U → W



Node	Heuristic Value
A	10
B	8
C	6
D	9
E	4
F	5
G	0
H	3

Goal Node: G
Output:- Start Node: S
 Goal Node: G

Node Expanded: S
 Node Expanded: B
 Node Expanded: E
 Node Expanded: G

Goal Node Found!
Optimal Path:
 S → B → E → G